Andrew McKnight

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Education

Old Dominion University · M.Sc., Dept. of Computer Science **2013**

- Thesis-track research in bioinformatics, robotics and 3D image meshing applied to protein folding
- Three published papers with two conference presentations, winning multiple awards
- C++ programming teaching and lab assistant for three semesters

Old Dominion University · B.Sc., Computer Science **2012**

- Graduate-level versions of courses as part of the Accelerated Master's program
- Minor in actuarial mathematics, up to advanced probability, diff-eq and multivariate calculus

Experience

Sentry · Senior Software Engineer · Remote **Jan 2021 – Current**

- *Stack*: Objective-C++, Swift, Python, SwiftUI, Django, Claude Code, SPM, XcodeGen, CocoaPods, Carthage, Swiftlint, clang-format, CodeCov, pre-commit, Fastlane, Arkana, GitHub Actions, Redash, SauceLabs, TestFlight, Linear
- *Deliverables*
  - Sentry Cocoa SDK: mobile app crash and performance monitoring
    - Integrated Specto's profiler into Sentry's Cocoa SDK and iterated with new API/features
    - Built User Feedback widget, the first user-facing UI shipped in Sentry Cocoa SDK
    - Built demo apps for SDK and Sales teams flexible enough to simulate any customer configuration
    - Parallelized long-running CI jobs and modularized test suites to find and fix statefulness issues
  - Seer for iOS and macOS: proposed and built native apps to manage our agentic offerings
  - Billing: platformized legacy system for product teams to self-serve how they charge for features

Specto · Senior Software Engineer · Remote **Nov 2019 – Jan 2021**

- *Stack*: Objective-C++, Swift, Go, Python, Protobuf, Jinja, XcodeGen, Bazel, pre-commit, GitHub Actions, Google Cloud Run
- *Highlight*: First employee at startup and remained through acquisition by Sentry
- *Deliverable*: Specto SDK, mobile app profiling SDK for iOS and Android
  - Shared C++ core between iOS and Android with Protobuf RPC to Python/Go backend
  - Python scripts to generate scoped Xcode projects from Bazel targets (pre `rules-xcode`)
  - Hardened SDK init sequence, app launch profiling, NSURLProtocol hooks, MetricKit ingestion
  - Signal and mach exception propagation with open source contributions to other crash reporters
  - Smoke test generation validating integration across CocoaPods, Carthage, Bazel and SPM

Rightpoint · Senior Software Engineer · Remote · Contract-to-Perm **Jan – Oct 2019**

- *Stack*: Swift, SwiftGen, SwiftLint, Sourcery, CircleCI, Postman, Browserstack, Countly
- *Deliverable*: Security app and SDK (under NDA)

Insight Therapeutics · Lead iOS Developer · Remote · Contract **Nov 2011 – Jun 2019**

- *Stack*: Objective-C, Swift, CoreGraphics, CoreData, MessageUI, UICollectionView, FMDB, SQLite, Storyboards, SSZipArchive, RNCryptor, cocoapods-keys, Trello, BitBucket, AWS
- *Highlight*: Sole consultation and development of multiple apps shipped in App Store and internally
- *Deliverables*
  - ClotRx: public health reference e-pamphlet about anticoagulants
    - Preexisting hybrid app with webviews and a PDF viewer, modernized by me
    - Migrated UIWebView→WKWebView, added search and bookmarking, redesigned for dark mode
    - Built interactive clinical patient eval modules with print/email reporting capability
  - DementiaRx: public health reference e-pamphlet about dementia
    - First new app created by me shipped to the App Store for a contracting engagement

Insight Therapeutics continued on next page

  - Scripted automatic content extraction from original paper publication PDFs
  - Built search, bookmarking and interactive patient eval elements later backported to ClotRx
- Antibiotic Kinetics: antibiotic dosage calculator
  - Ported from partner developer's Android app codebase, including adding first tests
  - Improved UX for error reporting through multistep calculations and data entry
- eConsent: HIPAA-compliant informed consent app for clinical trial participants
  - Interactive quiz-based presentation of study details with signature collection and export
  - Secure admin mode for saving/resuming progress and managing completed consents
  - Internal deployment using enterprise signing to hundreds of facilities

121nexus · Software Engineer · Remote · Contract **Jan – Apr 2018**

- *Stack*: Swift, AVFoundation, OpenCV, Firebase, cocoapods-keys, AWS
- *Deliverable*: 121 Scan, medical inventory management app/SDK with custom barcode/QR scanning
  - Public launches via iTunes and a white-labeled enterprise deployment in logistics facilities
  - Optimized critical barcode scanning component for high throughput pass/fail operation
  - Traced silent scan failures to an AVFoundation defect with double-FNC1 GS1 barcodes
  - Added crash reporting (Crashlytics), analytics (Firebase) and logging (XCGLogging)

Layer · Software Engineer · Remote **Dec 2016 – Jul 2017**

- *Stack*: C, Objective-C, Swift, OCMock, Specta, Expecta, Rails, Thrift, WebSockets, SPDY, SQLite, CocoaPods, Jenkins
- *Deliverable*: four-layer SDK offering for data→widgets→prebuilt module→sample app
  - Separated unit, integration and end-to-end tests to tighten development iteration loops
  - Streamlined developer environment configuration (Rake, Gems, CocoaPods and git submodules)
  - Migrated network protocol with Erlang backend from SPDY to WebSockets under Thrift RPC
  - Rearchitected auth and transport layers as state machines using TransitionKit

Twitter · Software Engineer · Cambridge, MA **Apr 2015 – Dec 2016**

- *Stack*: Objective-C, C, Swift, Java, Scala, Clojure, Coffeescript, Rails, React, Protobuf, Storm, Heron, Cascading, Scalding, Graphite, Beanstalk, Kafka, Redis, Cassandra, AWS, TeamCity, Chef, Trello, Vertica, Zeppelin
- *Highlights*
  - Presented NSOperation rearchitectures and open source tools at MobileDevCon 2016 in Amsterdam
  - Work on high-volume lambda backend, including on-call and migration from Storm to Heron
- *Deliverables*
  - Crashlytics SDK: crash reporting and beta deployment SDK for Apple platforms and Android
    - Rearchitected CLIs to bootstrap projects, upload dSYMs and interop with our macOS app
    - Added Bitcode and tvOS compatibility to SDKs, and investigated watchOS support
    - Researched and built feature surfacing solutions to detectable ecosystem-wide crash causes
  - Fabric SDK and macOS app: developer tool ecosystem management
    - Streamlined build and release processes with TeamCity agents deployed via Chef
    - Rearchitected macOS app launch sequence using queues of async/compound NSOperations
  - FABConfig: standardizes config of all Xcode projects and released the open source FABConfig
  - FABOperation: custom NSOperation subclasses for async work and grouping tasks

MC10 · Software Engineer · Cambridge, MA **Jan – Apr 2015**

- *Stack*: Objective-C, Swift, CoreBluetooth, NSOperation, TeamCity, Fabric
- *Highlights*
  - Rearchitected SDK and app codebases with NSOperation queues managing BTLE link
  - Drove adoption of Agile practices, working directly with PMs and via company-wide presentations
  - Introduced engineering practices via Git repos, GitHub and CI/CD pipelines with TeamCity/Fabric
- *Deliverable*: iOS SDK and companion app with BTLE link to proprietary medical wearable hardware

Raizlabs · Software Engineer · Boston, MA **Jul 2013 – Dec 2014**

- *Stack*: Objective-C, Swift, EventKit, KVO, Mustache, Core Data, QuickLook, MessageUI, WKWebView, Jenkins, TeamCity, Crittercism, Appboy, Urban Airship, CoreMetrics, Uncrustify
- *Deliverables*
  - Care.com: service seeker/provider marketplace
    - iOS 7 redesign, with custom perspective zoom effect in UIPageController with CATransform3D
    - Added floating quick-post button, swipe UITableViewRow actions, UIDataDetectorTypes usage
    - Improved pull-to-refresh, data entry with inputAccessoryView, UIAlertView presentation/routing
  - B&H Photo: e-commerce app for a high end brick-and-mortar merchant
    - Helped rebuild B&H's iPhone app on the new iPad codebase as a universal iOS 7 app
    - Push notifications and deep linking via Appboy, crash reporting and breadcrumbs via Crittercism
    - Custom parallactic scrolling of table view headers, and barcode scanning for product search
    - Custom UICollectionView layouts and animated transitions between grid and row layouts
  - UniFirst: interactive enterprise sales catalog on iPad
    - Improved drag-and-drop clothes modeler and emblem designer built with Auto Layout and KVO
    - Generated sales proposal PDFs from Mustache HTML templates, with model/emblem snapshots
    - Added Core Data model deep copying and orphan cleanup, enabling editing existing sales orders
  - Invaluable: realtime luxury auction platform on iPad
    - Refactored critical bidding calculation routines and corrected mathematical errors
    - Added pull-to-request footer in infinite scroll UITableView, horizontal UICollectionView scroll
    - Fixed statefulness issues with UI elements dependent due to race conditions in server links

ODU Systems Lab · Software Engineer · Norfolk, VA · Intern **Oct 2012 – May 2013**

- *Highlight*: Supervised two international students and led consulting meetings with clients on site
- *Deliverables*
  - ChirpXM: location based social networking app based on zip code centroid radius
  - Prototyped EMR iPad app for CHKD, with basic demographic and health chart input and display

Intergraph · Software Engineer · Hampton, VA · Intern **May – Oct 2012**

- *Stack*: C#, VisualBasic, SQL, QF-Test, Coverity, Visual Studio, TeamFoundation
- *Deliverable*: SmartMarine/SmartPlant3D, CAD system for designing ships and power plants, with multilayer blueprint export
  - Unit and UI testing, numerical modules baseline regression testing, and montecarlo simulation
  - Manual acceptance testing for new features and fixes; database migrations with handwritten SQL

Old Dominion University · Project co-lead · Norfolk, VA · Capstone **Aug 2012 – May 2013**

- *Stack*: Objective-C, OpenEars, MapQuest
- *Highlights*
  - Co-lead of winning seven-member team (out of four total) and sole developer of our iOS app
  - Four presentations given to visiting professors and industry executives, including demos and Q&A

Old Dominion University · Research Assistant · Norfolk, VA **Aug 2012 – May 2014**

- *Stack*: Java, C++, CGAL, Boost, Jama
- *Highlights*
  - Image processing, robotics and machine learning applications in bioinformatics
  - Multiple publications and conference presentations, winning multiple awards

Old Dominion University · Lab Instructor · Norfolk, VA **Aug 2012 – May 2014**

- Online delivery of CS 333: Problem solving and programming in C++
- Lab instructor for CS 250: Intermediate programming in C++
- Assistant instructor for CS 150: Introduction to programming in C++
- Website maintenance, grading, office hours

XVD Longboards · Web Developer · Norfolk, VA · Contract **Oct 2010 – Apr 2011**

- *Stack*: ActionScript, Flash Studio
- *Highlight*: Paid with a couple custom longboards, which I got to help build myself!
- *Deliverable*: Flash website with longboard designer applying custom graphics to multiple board shapes

Publications

- *Protein Loop Length Estimation from Medium Resolution CryoEM Images (M.Sc. thesis)* · Mirror: [1]
- *Estimating loop length from CryoEM images at medium resolutions* · Mirror: [1]
- *Estimating Lower Bounds on the Length of Protein Polymer Chain Segments using Robot Motion Planning* · Mirror: [1]
- *CryoEM Skeleton Length Estimation using a Decimated Curve* · Mirrors: [1] [2]

Honors & Awards

- **2013** · **Gene Newman Award** · Virginia Modeling, Analysis and Simulation Center Capstone conference · Best overall paper
- **2012** · **Graduate with Distinction** · Old Dominion University · Awarded with B.Sc.
- **2012** · **Undergraduate Research Award** · ODU Spring Undergraduate Research Fair · Outstanding undergraduate achievement
- **2011** · **First Place** · ODU-ACM Programming Contest
- **2010 – 2011** · **Dean's List** · Old Dominion University
- **2009** · **Summa Cum Laude** · Piedmont Virginia Community College

Presentations

iOSDevCampCO · Colorado Springs, CO **May 2018**

Colloquial Swift (45 min)

- Presented findings from analysis of common function implementations in open source repositories as a way to generate possible Swift Evolution proposals

CocoaHeads Boston · MIT, Cambridge, MA **Jun 2017**

UIViewController Decomposition and UI Reuse (20 min)

- Discussing strategies to separate responsibilities from UIViewControllers and a library of reusable UI components, e.g. animated transparent modal presentation with a close button

Boston Algorithms Meetup · Twitter, Cambridge, MA **Jan 2017**

HyperLogLog (20 min)

- Helped organized this initial meeting of the group
- Covered the theoretical workings of the HyperLogLog algorithm, with a copresenter giving a demo

mDevCon · Amsterdam, NL **Mar 2016**

Command and NSOperation (45 min)

- Delivered presentation on the Command software design pattern, and NSOperation
- Introduced subclasses we open sourced at Crashlytics for asynchronous and compound operations
- Previously delivered at a lunchtime tech talk at Twitter, Cambridge, MA (Nov 2015) and CocoaHeads Boston meetup, MIT, Cambridge, MA (Dec 2015)

VMASC Capstone Conference · Suffolk, VA **Apr 2013**

Estimating lower bounds on the length of protein polymer chain segments using robot motion planning

- Best paper award on research using motion planning algorithms to model protein molecule structures

IEEE WCSB · Philadelphia, PA **Oct 2012**

CryoEM skeleton length estimation using a decimated curve

- Using a dynamic graph algorithm to predict sequences of protein molecule substructures from 3D images

ODU Spring Undergraduate Research Symposium · Norfolk, VA **Apr 2012**
CryoEM skeleton length estimation using a decimated curve
- Developed further as a submission that was accepted to the IEEE WCSB later that fall

Mentorship

Girls Who Code · Twitter, Cambridge, MA **Jul 2016**
iOS Dev Session
- Part of GWC summer internship covering programming first principles to full-stack style development
- Designed and led iOS session where we built a Flappy Bird clone, introducing them to Xcode as an IDE, Interface Builder, Playgrounds and Swift

NeighborNest · Twitter, San Francisco, CA **Oct 2015**
Web Dev Session
- Community program to help advocate technology literacy
- Assisted in session where students worked on a Twitter clone, teaching functional programming concepts, Javascript, and some front- and back- end concepts